Vector Autoregressive Models (VARs)

These notes accompany Lecture 13. They combine the slide material with the explanations given in class. Terms printed in bold are defined in the glossary at the end.

A VAR is the workhorse of applied macroeconomic time-series analysis: cheap to estimate, and – once identification is added – able to answer causal questions. The running example is a small monetary model of the euro area, estimated on quarterly data.

Learning outcomes – after this lecture you should be able to:

1. Write down a VAR(p), state the assumptions on the errors, and explain why OLS equation-by-equation is appropriate.
2. Check stationarity and choose a lag length, balancing over-fitting against omitted variable bias.
3. Run and interpret VAR residual diagnostics, and know which failures are serious.
4. Produce deterministic and stochastic forecasts and build alternative scenarios.
5. Explain why identification is needed, apply and defend a recursive ordering.
6. Read an impulse response function and a forecast-error variance decomposition.

Notation used throughout

Symbol	Meaning
y_t	The n × 1 vector of variables at time t
a_0 , $A_1 \dots A_p$	Reduced-form intercepts and lag coefficient matrices, with entries a_{jk}
e_t , Σ_e	Reduced-form errors (one-step-ahead forecast errors) and their covariance matrix
B , b_{jk}	Structural contemporaneous matrix, unit diagonal, with off-diagonal entries b_{jk}
Γ_0 , $\Gamma_1 \dots \Gamma_p$	Structural intercepts and lag coefficient matrices, with entries γ_{jk}
u_t , D	Structural shocks and their (diagonal) covariance matrix
Ψ_i , Θ_i	Reduced-form and structural moving-average coefficients, with $\Theta_i = \Psi_i B^{-1}$
Γ_y(h)	Autocovariance matrix of y_t at lag h
P , S	Cholesky factor of Σ_e and the diagonal matrix of structural standard deviations, $P = B^{-1}S$

Notation follows Enders (2015, ch. 5). Lütkepohl (2005) writes the reduced-form lag matrices as A_i likewise but uses v for the intercept vector; Kilian and Lütkepohl (2017) write the structural contemporaneous matrix as B_0. Check which convention is in use before comparing formulae across textbooks.

Contents

1. Introduction to VARs
2. Stationarity and lag specification
3. Residual diagnostics
4. Forecasting with VARs
5. Structural VARs
6. Impulse responses and variance decomposition

1. Introduction to VARs

1.1 Sims' critique

It is common in economics to have models in which some variables are not only explanatory variables for a given dependent variable but are also explained by the variables they help determine. These are simultaneous-equations models, and using them requires deciding which variables are endogenous and which exogenous.

Sims (1980) argued that in the large-scale macro-econometric models of the 1960s and 1970s that decision was doing far too much work. Identification rested on declaring most variables exogenous and most cross-equation channels absent – restrictions that were asserted rather than tested, and which he did not find credible.

The consequence is radical: there should be no distinction between endogenous and exogenous variables. Once the distinction is abandoned, all variables are treated as endogenous.

The trade is explicit. You give up a prior story about which variable causes which; in exchange the dynamics come from the data rather than from the modeller. You get the causal story back only in Section 5, and only by stating an assumption you are prepared to defend.

Sims, C. A. (1980). Macroeconomics and Reality. Econometrica, 48(1), 1–48.

1.2 What VARs are and what they are for

A VAR is a system of n equations in n variables. Each equation regresses one variable on p lags of every variable in the system, including its own. Every variable is endogenous: there is no left-hand and right-hand side to the system as a whole, only to each equation.

Purpose	What you need	Why
Forecasting	A reduced-form VAR is sufficient	Estimation alone. VARs frequently forecast better than univariate models because they exploit cross-variable dynamics
Structural / policy analysis	A structural VAR	Estimation and identification. Needed to trace responses to shocks, attribute business-cycle fluctuations to sources, and discriminate between competing theories

1.3 The model

A bivariate VAR with one lag makes the structure visible. Each series is explained by the common past of both:

$$y1,t = a10 + a11y1,t−1 + a12y2,t−1 + e1,t$$

$$y2,t = a20 + a21y1,t−1 + a22y2,t−1 + e2,t$$

Or compactly, $y_t = a_0 + A_1 y_{t-1} + e_t$, where a_0 is a vector of intercepts, A_1 a matrix of coefficients and e_t a vector of reduced-form errors. A VAR(p) generalises a univariate AR(p) to n variables, and carries $n + pn^2$ coefficients before the error covariances are counted.

The assumptions on the errors – read these carefully

$E[e_t] = 0$, and the errors have finite variance.

The error terms of the individual equations MAY be contemporaneously correlated – the covariance matrix Σ_e is not assumed to be diagonal.

But the errors must be uncorrelated with their own lagged values and with all the right-hand-side variables, that is, the lagged y's.

That middle point is the whole reason Section 5 exists. Because the errors are not orthogonal to one another, they cannot be interpreted as fundamental or structural shocks. They are one-step-ahead forecast errors, and the word "shock" is best reserved until Section 5.

A VAR with no economic restrictions imposed is called a reduced-form VAR, as opposed to a structural VAR.

1.4 Estimation

Estimation is by OLS, applied one equation at a time. Two properties follow:

- **Consistent,** because only lagged values of the endogenous variables appear on the right-hand side. Since e_t is uncorrelated with all past errors, it is uncorrelated with anything built from them – including every lagged y on the right.
- **Asymptotically efficient,** because all equations have identical regressors, so system estimation by GLS returns numerically identical estimates – there is no efficiency gain left for SUR to capture. This holds only if the lag order is right: if it is not, the errors are autocorrelated and the estimates are consistent for a model nobody wanted, which is why Section 3 exists.

Specification choices

- **Which variables** – driven by the question, by theory, and by what the sample will bear.
- **What else on the right-hand side** – a constant, a time trend, dummies, genuinely exogenous series.
- **Which transformation** – levels, gaps, log differences or growth rates. Levels retain long-run information but need stationarity handled elsewhere; differences guarantee stationarity and discard the long run.
- **Parsimony** – not a preference but a constraint; see Section 2.4.

The worked example uses quarterly euro area data from 2002Q1, with four endogenous variables – the output gap (HP-filtered log GDP), CPI inflation, the monetary policy rate, and the real effective exchange rate gap (HP-filtered log REER) – two lags, and a constant. R^2 values in VARs are typically high, because the models include many lags and therefore a great deal of information.

2. Stationarity and lag specification

2.1 Covariance stationarity

The conditions mirror the univariate case. A VAR is covariance-stationary when two things hold:

- The mean vector $\mu = E[y_t]$ is the same at every t – μ collects the individual means of the processes.
- The autocovariance matrices $\Gamma_y(h) = E[(y_t - \mu)(y_{t+h} - \mu)']$ depend on the distance h only, and not on where in the sample you are standing.

Note what the second condition does not say. $\Gamma_y(h)$ is not symmetric for $h \neq 0$: the covariance of y_1 today with y_2 next quarter differs from the covariance of y_2 today with y_1 next quarter. That asymmetry is precisely the dynamic information a VAR exists to capture, and it has no counterpart in the univariate case.

If a VAR is covariance-stationary, it implies that the response to innovations gradually dies down, tending to zero in the long run – which is exactly what makes an impulse response readable as a temporary deviation rather than a permanent shift.

2.2 The VMA(∞) representation

If a VAR is stationary, inverting the lag polynomial writes y_t as an infinite sum of past errors:

$$yt = \mu + et + \Psi 1et-1 + \Psi 2et-2 + \ldots = \mu + \Sigma i=0..\infty\ \Psi iet-i\ ,\quad with\ \Psi 0 = In$$

This is the vector moving average representation, also known as the Wold representation or Wold decomposition. It is the same logic as writing a stationary AR process as an infinite-order MA. Read the Ψ matrices as coefficients: Ψ_i measures how much a one-unit reduced-form error i periods ago still moves y_t today. Stationarity is what makes the sum converge – without it there is nothing to plot.

These are not yet impulse responses. Ψ_i traces reduced-form errors, which are correlated with one another, so "a one-unit change in e_{1t} holding e_{2t} fixed" describes nothing that happens in the data. Section 6.1 returns to this: the structural coefficients $\Theta_i = \Psi_i B^{-1}$ are the ones that can be plotted, and identification is exactly what converts one into the other.

2.3 Checking stationarity in EViews

View → Lag Structure → AR Roots Table or Graph. The estimated VAR is stationary if all the inverse roots of the characteristic AR polynomial lie inside the unit circle. Note the inversion: EViews plots inverse roots, so "inside" is the stable case – read the axis label, not the picture.

This matters practically: if the VAR is not stable, certain results such as impulse response standard errors are not valid. If the characteristic polynomial has a root equal to one, then some or all of the variables in the VAR are integrated of order one.

2.4 How many lags?

	Consequence
Too short	Omitted dynamics land in the errors, which then fail the white-noise requirement – and everything computed downstream (impulse responses, forecast bands, decompositions) is derived as though that requirement held
Too long	$n + pn^2$ coefficients against T observations. In-sample fit improves by construction; the estimates get noisier and the forecasts get worse

	Consequence
Enough	The residuals pass the autocorrelation test at the horizons you care about. That test, not the information criterion, is the criterion

The multidimensional versions of the familiar information criteria apply: AIC, the Schwarz criterion (SC), and the Hannan-Quinn criterion (HQ). All trade parsimony against fit, but they are not interchangeable: SC and HQ are consistent for the true lag order, while AIC is not and tends to over-select in finite samples. They disagree routinely, and when they do, estimate at the competing orders and let the residual diagnostics decide.

> **Rules of thumb**
>
> With T observations, n variables and p lags you are estimating $n + pn^2$ coefficients from $n(T - p)$ observations. On quarterly data that usually leaves $p = 4$ as a ceiling for anything but the smallest systems; on monthly data $p = 12$ is common, but only for two or three variables.
>
> When testing in EViews, set a maximum of 12 or 24 lags for monthly data, 4 or 8 for quarterly, 1 or 2 for annual.
>
> Practitioners keep $n \cdot p$ well under $T/3$, where n is the number of variables and T the sample size.
>
> Note: if there are no exogenous variables the lag count starts at one; with a constant selected it starts at zero.

One caveat: if you use a VAR in long-run levels form – possible when the variables are I(1) and cointegrated – the requirement that the residuals be white noise is considered less important. But it is generally better to estimate VARs on stationary variables, and to use error correction models when the variables are cointegrated.

2.5 Over-fitting versus omitted variable bias

Estimating a VAR means facing a trade-off:

- **Over-fitting.** Too many coefficients chase too little data, so the coefficients themselves are poorly estimated. The model appears to fit the in-sample data very well, but out-of-sample forecasts are bad.
- **Omitted variable bias.** Trimming variables or lags to save degrees of freedom can leave out important variables or dynamics, biasing the estimates.

Three responses, in increasing order of machinery. A core plus rotation: estimate a small core VAR of key macroeconomic variables and cycle the others through in small groups – standard practice, which manages the problem rather than solving it. Shrinkage: Bayesian VARs with a Minnesota-type prior pull the coefficients toward a random walk, letting you carry more variables than OLS would tolerate, because the prior does the work that degrees of freedom cannot. Factors: replace many series with a few principal components, the FAVAR approach.

3. Residual diagnostics

For a stationary VAR to be correctly specified, the residuals need to be white noise. Start by plotting them (View → Residuals).

In the euro area example they are not. The single most visible feature is the output gap outlier in 2020 – the COVID lockdowns and the rapid rebound. Inflation and the policy rate both show a cluster of large residuals in 2021–2024, the post-pandemic inflation surge and the ECB tightening cycle. So there are two distinct problems: a COVID outlier and a 2021–24 inflation-policy episode.

The residual correlation matrix (View → Residuals → Correlation Matrix) shows non-zero off-diagonal elements – the residuals are correlated among themselves. This is expected and not a defect: it is a reduced-form VAR with no structural identification imposed.

3.1 Correlograms

> **Reading the correlogram grid**
>
> The diagonal panels are the residual autocorrelations. These are the ones that bear on the white-noise requirement.
>
> The off-diagonal panels are cross-lag correlations between different equations' residuals. These are NOT what white noise requires you to eliminate.
>
> The dotted lines are ±2 standard error bounds, i.e. the 5% level.

3.2 The autocorrelation LM test

Absence of residual autocorrelation is the most important requirement. The LM test checks the null of no serial correlation up to a chosen lag order, and is general enough to apply to VARs in levels.

> **Why failing this test is the serious case**
>
> Leftover autocorrelation means the lag length is too short, or the dynamics are misspecified.
>
> The estimates then remain consistent only for the model you actually fitted – they do not recover the dynamics you are after.
>
> And the impulse-response standard errors and variance decompositions become invalid.
>
> The fix is almost always to add lags and re-test.

EViews reports two panels answering different questions:

Panel	Question	Use
No serial correlation at lag h	Is there autocorrelation specifically at lag h, holding the others aside?	Diagnostic – tells you where the problem sits
No serial correlation at lags 1 to h	Is there any autocorrelation anywhere up to lag h?	The conservative overall verdict

In the example the partial panel shows two clear spikes, at lag 3 ($p = 0.0011$) and lag 7 ($p = 0.0006$). In the cumulative panel lag 1 alone is fine ($p = 0.39$), but from lag 2 onward the null is rejected at every horizon. The overall conclusion: the VAR fails the no-serial-correlation null, and the residuals are not white noise.

3.3 Homoskedasticity and normality

Test	What failing it costs
White heteroskedasticity test	Does not break consistency – point estimates are still fine. What it damages is inference: standard errors are wrong, so t-statistics, confidence bands and IRF error bands are unreliable. For macro data with crisis episodes this test frequently fails, which is usually acceptable provided you use robust inference for the IRFs (bootstrapped standard errors)
Normality (Jarque-Bera type)	Normality is not needed for consistency; by asymptotic theory the estimators and tests remain valid in large samples without it. In practice macro residuals are almost always non-normal

Note the hierarchy: autocorrelation is the failure that must be fixed, because it means you have estimated the wrong model. Heteroskedasticity and non-normality damage inference only, and can be handled with robust methods.

3.4 Fixing the problems

1. Add impulse dummies for the pandemic. Dummies for 2020Q1 and Q2 account for the lockdowns; Q3 and Q4 account for the post-COVID recovery.
2. Add a second lag of the endogenous variables to capture the dynamic structure more accurately and mitigate the autocorrelation.
3. Add changes in oil prices as an exogenous variable – a significant factor in euro area inflation but external to the system.

Follow these steps sequentially, checking the results at each stage. After the fixes the residuals look much more like white noise, the autocorrelation problem is eliminated, and the VAR remains stationary.

4. Forecasting with VARs

An estimated VAR forecasts the same way a univariate AR does: iterate the estimated equations forward, substituting forecasts for the lagged values that are not yet observed. Forecasts are easiest to produce through a Model object – from the estimated VAR, Proc → Make Model.

4.1 Deterministic versus stochastic

The model window then offers Solve, which opens the model solution dialog. Because a VAR forecast is iterative, beyond the first period the lagged endogenous values are themselves forecasts; any exogenous series must therefore be extended over the whole forecast horizon before you solve. This is the step most often forgotten, and the model will simply refuse to solve without it.

A deterministic forecast assumes, in effect, that the stochastic equations hold exactly over the forecast period. That ignores two things: random disturbances, and the fact that the coefficients are estimated rather than known.

A stochastic simulation accounts for both, producing error bounds. In EViews: from the model window choose Solve, select stochastic instead of deterministic, and under Solution Scenarios and Output choose either standard deviations or bounds (95% intervals).

EViews adds series with the suffixes _0m (mean forecast), _0l (lower bound) and _0h (upper bound), where the 0 denotes the baseline scenario. In the example the actuals remain within the confidence bands, which suggests the model is doing reasonably well once uncertainty is acknowledged.

4.2 Alternative scenarios

1. Open the model, click Scenarios, and create a new scenario so as not to overwrite the baseline.
2. Solve the existing model under the new scenario first – this must be done before overriding any variable.
3. Click Variables, right-click the variable whose path you want to change, and select Edit → Override.
4. Enter the changes as a delta (an absolute change) or a delta percent (a percentage change) relative to the unadjusted forecast.
5. Solve again. EViews adds series with the suffix _1 for the new scenario, which can then be graphed against the baseline and the actuals.

> **A result worth pausing on**
>
> In the worked scenario the policy rate is cut more aggressively than in the baseline – and the output gap comes out slightly LOWER, which is the opposite of what theory would suggest.
>
> This is not a bug. It reflects the historical behaviour of the data: in this sample, lower interest rates are usually associated with a weaker economy, because monetary policy reacts to disinflationary pressure by cutting.
>
> The reduced-form VAR has learned the central bank's reaction function, not the causal effect of policy. This is precisely why Section 5 is necessary.

5. Structural VARs

5.1 Why identification is needed

Consider this sequence. The central bank anticipates a rise in inflation and raises the policy rate. Inflation still rises for some time, as anticipated.

The wrong conclusion is that the rate hike caused inflation to rise. What actually happened is that the monetary contraction was endogenous – policy reacted to expected inflation.

The problem is general. Policy effects cannot be measured while the policy variable is itself responding to everything else in the system. The reduced-form error on the rate equation is a forecast error, not a policy shock: it bundles the part of the rate move that responded to conditions the bank observed together with the part that was genuinely unanticipated, and only the second is a shock in the sense we need. What we want is the component that responded to nothing, and the path the economy takes after it. Recovering that from the estimated residuals is what identification means.

5.2 The algebra

Suppose a structural VAR(1) with mutually uncorrelated structural shocks u, in two variables – inflation π and the policy rate r:

$$B\,y_t = \Gamma 0 + \Gamma 1 y_{t-1} + u_t\,, \quad \text{with} \quad E[u_t u_t'] = D \text{ diagonal}$$

Matrix B holds the contemporaneous relations between the endogenous variables; each off-diagonal entry is a claim about what moves what within the period, and the unit diagonal is a normalisation. The system cannot be estimated as it stands, because y_t appears on both sides of both equations, so OLS on either one has an endogenous regressor. Pre-multiplying by B^{-1} removes the simultaneity and gives the reduced-form VAR we can actually estimate:

$$y_t = a0 + A1 y_{t-1} + e_t\,, \quad \text{where} \quad a0 = B^{-1}\Gamma 0\,,\ A1 = B^{-1}\Gamma 1 \ \text{and} \ e_t = B^{-1}u_t$$

So the forecast errors are linear combinations of the structural shocks: each one is its own structural shock contaminated by the other. A non-diagonal Σ_e is therefore not a nuisance – it is the contemporaneous structure, showing up in the only place the reduced form can put it, since $\Sigma_e = B^{-1} D\, B^{-1\prime}$.

> **The intuition**
>
> When inflation surprises us, we usually do not know why it happened. Monetary, fiscal, external and cost factors could all be contributing.
>
> The reduced-form residual is that mixture. The structural shocks are the separate ingredients, and identification is the recipe for separating them.
>
> The structural VAR cannot be estimated directly – it is a theoretical construct and unobservable. What we have is time series, from which we estimate a reduced-form VAR and recover coefficients and residuals.

5.3 Counting the restrictions

For a two-variable VAR(1):

	Parameters	Breakdown
Reduced-form VAR (known)	9	6 coefficients (2 intercepts and 4 lag coefficients), plus 2 variances and 1 covariance from Σ_e
Structural VAR (unknown)	10	6 intercepts and lag coefficients, 2 contemporaneous parameters (b_{12} and b_{21}), and 2 structural variances – the covariance is zero by assumption, since the structural shocks are uncorrelated
Shortfall	1	One restriction must be imposed

The gap is not a small-sample problem, and no amount of extra data closes it. In general, B carries $n^2 - n$ unknown off-diagonal elements plus n structural variances, so n^2 unknowns in all, while the symmetric Σ_e yields $n(n+1)/2$ distinct numbers. The number of restrictions needed is the difference:

$$n(n-1)/2$$

So a three-variable VAR needs 3 restrictions and a four-variable VAR needs 6. Note that a scheme supplying exactly this many is exactly identified, and therefore produces no testable restriction – nothing in the diagnostics validates the choice. That is the honest limitation of the whole exercise.

5.4 Recursive ordering (Cholesky)

The usual approach is to impose zeros on matrix B, which is equivalent to imposing restrictions on the contemporaneous relations among the endogenous variables – a matter of economics rather than statistics.

Setting $b_{12} = 0$ means that π is affected by a shock to r only with a lag, while a shock to π affects r contemporaneously. If r is the monetary policy rate, this says the authorities observe inflation and react to it within the period, but their actions reach the economy only with a delay. With this restriction the number of unknowns equals the number of knowns, and the structural shocks can be recovered.

The resulting B^{-1} is lower triangular. A recursive ordering, in general, is a set of $n(n-1)/2$ zeros placed so as to make B triangular: economically, a sequential chain in which each variable may respond immediately to those before it and only with a lag to those after. Extending to three variables (y, π, r) with the rate last means the policy rate affects the other two only with a lag.

B^{-1} and the Cholesky factor are not quite the same matrix. B^{-1} has ones on the diagonal, whereas what software reports is the Cholesky factor P of Σ_e, which is lower triangular with a positive diagonal: $P = B^{-1}S$, where S is the diagonal matrix of structural standard deviations, so that $\Sigma_e = PP'$. The two encode the same restrictions and differ only in the units of the shock – P absorbs the standard deviations. That is why the default impulse in EViews is a one-standard-deviation shock, and why rescaling to a one-percentage-point shock (Section 6.1) is a legitimate change of units rather than a fudge.

> **Ordering matters – usually**
>
> If the correlations between the reduced-form errors are low, the order is practically irrelevant. But correlations are usually strong, and then the order matters a great deal.
>
> Order from the "least endogenous" to the "most endogenous". In the euro area example: GDP first (at time t only its own shock moves it), then the exchange rate, then inflation, then the policy rate last – assuming the ECB is forward-looking and reacts within the quarter to shocks in output, the exchange rate and inflation, while monetary transmission lags mean the rate affects them only later.
>
> The real exchange rate in second place is the assumption most open to challenge, since an exchange rate is a fast-moving asset price that reacts to news immediately, and much of the literature places it last for that reason. The argument for placing it early here is that the real effective rate on quarterly data is dominated by its slowly-adjusting relative-price component. Because the assumption is arguable, it is worth re-estimating with the exchange rate ordered last and checking that the responses of interest are not sensitive to the swap.
>
> Deciding the ordering is a decision about economics, not statistics. It must be justified.

5.5 Other identification strategies

Strategy	What is restricted
Recursive (Cholesky)	Contemporaneous effects, forced into a triangular order
Short-run, non-recursive	Still zeros in matrix B, but not triangular (Blanchard-Perotti)
Long-run restrictions	The cumulative or permanent effect rather than the impact effect (Blanchard-Quah)
Sign restrictions	The sign of the impulse responses for a few periods, rather than setting anything to zero (Uhlig)

6. Impulse responses and variance decomposition

Once identification has given us clean, independent structural shocks, we can ask two different questions.

6.1 The impulse response

The thought experiment: give the system a single one-time poke – one structural shock of size +1 – hold all other shocks at zero, and trace how every variable reacts this quarter, next quarter and on into the future. The reaction spreads over time because each variable depends on the lagged values of all the others.

Where IRFs come from

A stable VAR can be rewritten as the present plus an infinite echo of past structural shocks, with coefficient matrices Θ_0, Θ_1, Θ_2, … These are the Ψ matrices of Section 2.2 post-multiplied by the identification: $\Theta_i = \Psi_i B^{-1}$, and in particular $\Theta_0 = B^{-1}$, so identification is exactly what converts a reduced-form moving average into a structural one. The punchline is that those Θ matrices ARE the impulse responses: Θ_0 is the impact effect, Θ_1 the effect after one period, and so on – equivalently, $\text{IRF}(h) = \partial y_{t+h}/\partial u_t = \Theta_h$. Plotting them against the horizon simply draws the response curve.

Reading an IRF – four things to look for

What	Question
Sign at impact	Which way does the variable move right away?
Peak effect	How big does the effect get, and after how many quarters?
Persistence	How many quarters until the effect dies out and the variable returns to steady state?
Bands	The shaded region is the uncertainty. If the band includes zero, the effect is not distinguishable from nothing at that horizon

Always read the band, not just the line. And note a caveat on persistence: a return to zero indicates stability only for a VAR in stationary variables. If the VAR is in levels the response does not die out – instead the level of the variable eventually flattens.

Use bootstrapped error bands for macro data. They build confidence intervals by repeatedly re-sampling the model's actual residuals rather than assuming normality, which gives more reliable

bands when the errors are non-normal, fat-tailed or heteroskedastic – as real macro residuals usually are.

The euro area example

A contractionary monetary policy shock raises the rate, cools output, lowers inflation, and briefly appreciates the currency. Reading each response against its band:

Response	Reading
Rate → rate	Jumps about +0.15 on impact and decays to zero within roughly 8 quarters – a normal, persistent-but-temporary tightening. This is just the shock itself
Rate → output	Drifts negative and stays mildly below zero for most of the horizon – the textbook contractionary effect, though the band is close to zero, so economically sensible but only weakly significant
Rate → prices	Prices fall, growing to about −0.1 at long horizons with a band clearly below zero. No price puzzle – a rate hike lowers inflation, which is why you would trust this ordering
Rate → real exchange rate	Appreciates briefly, consistent with higher rates attracting capital. Small and short-lived

Because the VAR is linear, impulse responses scale proportionally to the shock size. To convert a response to a one-standard-deviation shock into a response to a one-percentage-point shock, divide every response in that column by the impact size of the shock – here about 0.15. This is the change of units described at the end of Section 5.4, not an adjustment to the model.

6.2 Forecast error variance decomposition

The IRF gives the shape of a shock's effect but not its importance relative to other shocks. To get at importance, think about forecasting: we do not know future shocks, so the forecast has error, and that error grows with the horizon. Variance decomposition splits that uncertainty into shares by shock.

At each horizon the forecast-error variance is sliced into the part coming from each structural shock, and the slices always add to 100%. It uses the very same Θ matrices as the IRF – squared and summed instead of plotted.

> **How to read it**
>
> Read across the horizon, not at impact. Under a recursive ordering the impact shares are fixed by the ordering itself – the first variable's first-period forecast error is entirely its own shock, by construction – so they tell you about your identifying assumption rather than about the economy.
>
> In the short run a variable's forecast error is mostly its own shock, because the other shocks have not had time to work through the system.
>
> As the horizon lengthens the other shocks take a larger share. In the euro area example, output uncertainty comes to be dominated by price shocks (~65%) and the policy rate largely by output shocks (~65%).
>
> That last figure carries the lecture's main message: most rate movements are an endogenous response to the economy, not exogenous policy shocks.

6.3 The two tools compared

	Impulse response	Variance decomposition
Question	What is the effect?	How much does it matter?
Shows	Direction and size of a shock's effect, and how it evolves over time	Relative importance of shocks; the sources of fluctuations
In one word	Reaction	Contribution

Both are built from the same identified shocks. The one thing to remember: no identification, no meaningful impulse response and no meaningful decomposition.

Key terms

Term	Meaning
Reduced-form VAR	Each variable regressed on lags of all variables, with no economic restrictions. Residuals are correlated across equations.
Structural VAR (SVAR)	A VAR with identifying restrictions imposed, so the shocks can be interpreted as fundamental.
Covariance-stationary VAR	Mean independent of time; covariance depending only on the lag between observations.
Inverse roots	Roots of the characteristic AR polynomial; all must lie inside the unit circle for stability.
VMA(∞) / Wold representation	y_t written as an infinite sum of past reduced-form errors, with coefficients Ψ_i; the source of impulse responses once identification converts them into $\Theta_i = \Psi_i B^{-1}$.
Hannan-Quinn criterion	One of the information criteria used for VAR lag selection, alongside AIC and Schwarz. Consistent for the true lag order, unlike AIC.
Autocorrelation LM test	Tests the null of no serial correlation up to a chosen lag. The serious diagnostic – failure means the model you fitted is not the one you intended.
Identification	Imposing restrictions on matrix B so the structural shocks can be recovered from reduced-form residuals. Exact identification needs $n(n-1)/2$ restrictions.
Recursive ordering / Cholesky	Triangular restrictions on contemporaneous effects; variables ordered least to most endogenous.
Ψ and Θ matrices	Reduced-form and structural moving-average coefficients, linked by $\Theta_i = \Psi_i B^{-1}$. The Θ_i are the impulse responses.
Cholesky factor (P)	The lower-triangular factor of Σ_e with positive diagonal, $P = B^{-1}S$, where S holds the structural standard deviations. The same restrictions as B^{-1}, but in one-standard-deviation shock units.
Sign restrictions	Identification by restricting the sign of impulse responses rather than setting coefficients to zero.
Impulse response function (IRF)	The dynamic path of every variable following a single structural shock.
Bootstrapped error bands	Confidence bands built by re-sampling actual residuals rather than assuming normality.

Term	Meaning
Forecast error variance decomposition	The share of a variable's forecast uncertainty at each horizon attributable to each structural shock.
Price puzzle	The counter-intuitive finding that a rate hike raises prices; usually a sign of poor identification.

Review questions

Use these to check your understanding before the seminar.

1. What exactly did Sims criticise about large-scale macro models, and what does a VAR do instead?
2. Write down a bivariate VAR(1). Which errors may be correlated and which may not? Why does that distinction matter later?
3. Why is OLS equation-by-equation both consistent for a reduced-form VAR?
4. State the two conditions for a VAR to be covariance-stationary, and explain why stationarity matters for impulse responses.
5. What is the VMA(∞) representation and what role do the Ψ matrices play? Why are they not yet impulse responses?
6. Your VAR has a characteristic root equal to one. What does this tell you about the variables?
7. Explain the trade-off in choosing p. What does the constraint $n \cdot p < T/3$ imply for a 5-variable VAR on 80 quarterly observations?
8. Distinguish over-fitting from omitted variable bias.
9. In a residual correlogram grid, which panels bear on the white-noise requirement and which do not?
10. Why is failing the autocorrelation LM test more serious than failing the White or normality tests? Be specific about what breaks.
11. The partial LM panel shows spikes at lags 3 and 7 while the cumulative panel rejects from lag 2 onward. What do you conclude, and what would you do first?
12. Why are non-zero off-diagonal elements in the residual correlation matrix expected rather than a defect?
13. Distinguish deterministic from stochastic forecasting. What two sources of uncertainty does the second account for?
14. Why must exogenous series be extended over the forecast horizon before a VAR model will solve?
15. In the alternative scenario a more aggressive rate cut produces a LOWER output gap. Explain why this happens and what it reveals about reduced-form VARs.
16. Set out the central bank example that motivates identification. What is the wrong conclusion and why is it wrong?
17. Show how $e_t = B^{-1}u_t$ arises, and explain in words what it means for the reduced-form residuals.

18. Count the known and unknown parameters for a 2-variable VAR(1). How many restrictions are needed, and why is the structural covariance zero?
19. How many restrictions does a 5-variable VAR require? Use the formula.
20. What does setting $b_{12} = 0$ mean economically when r is the policy rate?
21. When does the Cholesky ordering not matter much?
22. List the four things to look for when reading an IRF. Why must you read the band and not just the line?
23. Why are bootstrapped error bands preferred for macro data?
24. The rate shock has an impact size of 0.15. How would you rescale the responses to a one-percentage-point shock, and why is that legitimate?
25. Why should a variance decomposition be read across the horizon rather than at impact?
26. The policy rate's long-horizon forecast error is ~65% due to output shocks. What does this say about the nature of monetary policy?

References and further reading

Sims, C. A. (1980). Macroeconomics and Reality. Econometrica, 48(1), 1–48.

Lütkepohl, H. (2005). New Introduction to Multiple Time Series Analysis. Springer.

Enders, W. (2015). Applied Econometric Time Series, 4th ed. Wiley. Chapter 5 is the closest match to this lecture.

Kilian, L., & Lütkepohl, H. (2017). Structural Vector Autoregressive Analysis. Cambridge University Press.

Hamilton, J. D. (1994). Time Series Analysis. Princeton University Press. Chapters 10 and 11.

Brooks, C. (2019). Introductory Econometrics for Finance, 4th ed. Cambridge University Press. Chapter 7, at a gentler level.

Macroeconometric Forecasting. IMF online course. URL: https://www.edx.org/course/macroeconometric-forecasting-2.

Asteriou, D., Hall, S.G. Applied Econometrics – Chapter 15.